

MAT 344 HW03

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Problem 1. Let $I = [0, 1]$ and consider a smooth map $H: I \times I \rightarrow \Omega \subseteq \mathbb{C}$ where Ω is a good domain. We consider H as a homotopy of curves $\gamma_t: I \rightarrow \Omega$ where $\gamma_t(s) = H(t, s)$ for $s \in I$. We fix $z_0 \in \Omega$ and assume that $\gamma_t(0) = \gamma_t(1) = z_0$ so γ_t is a family of loops in Γ based at z_0 .

- (i) Let $\varepsilon = \min_{(t,s) \in I \times I} \{\text{distance}(H(t, s), \partial\Omega)\}$. Why is $\varepsilon > 0$?
- (ii) Prove there exists $\delta > 0$ such that if two points $p, q \in I \times I$ have $|p - q| < \delta$ then $|H(p) - H(q)| < \varepsilon/4$.
- (iii) Let ω be a 1-form with $d\omega = 0$. Show that the line integrals $\int_{\gamma_t} \omega$ are independent of $t \in [0, 1]$.

Solution. (i) The function $d: (I \times I) \times \partial\Omega \rightarrow [0, \infty)$ such that $d((t, s), p) = |H(t, s) - p|$ is continuous. Since $I \times I$ and Ω are compact, their product is compact, so d achieves a minimum value which is equal to ε by definition. Assuming that H maps $I \times I$ into the interior of Ω so that $H(t, s)$ doesn't lie on $\partial\Omega$ for any (t, s) , $d((t, s), p) > 0$ everywhere, so its minimum must be greater than zero.

(ii) This follows from the fact that $H: I \times I \rightarrow \Omega$ is continuous with compact domain, hence uniformly continuous.

(iii) Let n be a positive integer such that $1/n < \delta/2$. For each $j, k \in \{1, \dots, n\}$, the squares $R_{j,k} = [(j-1)/n, j/n] \times [(k-1)/n, k/n]$ are each contained in an open ball of radius $\varepsilon/4$ that is contained in the interior of Ω , which are in turn contained in squares $S_{j,k}$ of radius $\varepsilon/2$ with side lengths parallel to the axes that are still contained in the interior of Ω . That is, $S_{j,k}$ contains $H(R_{j,k})$.

For a fixed j and $k \in \{0, 1, \dots, n\}$, let $a_k = H((j-1)/n, k/n)$ and $b_k = H(j/n, k/n)$. Define $\gamma^{j,k}$ to be the restriction of $\gamma_{j/n}$ to $[(k-1)/n, k/n]$ and let γ^j be the path $\gamma_{j/n}$, for brevity. Let $\eta_k: I \rightarrow \mathbb{C}$ be the line segment going from a_k to b_k for $k \in \{0, 1, \dots, n\}$ and $\tilde{\eta}_k$ be the segment going in the other direction, from b_k to a_k . We argued in Tuesday's class that line integrals inside the $S_{j,k}$ depend only on the endpoints of the path being integrated over, so we can compute

$$\begin{aligned} \int_{\gamma^{j-1}} \omega &= \sum_{k=1}^n \int_{\gamma^{j-1,k}} \omega = \sum_{k=1}^n \left(\int_{\eta_{k-1}} \omega + \int_{\gamma^{j,k}} \omega + \int_{\tilde{\eta}_k} \omega \right) \\ &= \sum_{k=1}^n \left(\int_{\gamma^{j,k}} \omega + \int_{\tilde{\eta}_k} \omega + \int_{\eta_k} \omega \right) = \sum_{k=1}^n \left(\int_{\gamma^{j,k}} \omega \right) = \int_{\gamma^j} \omega \end{aligned}$$

hence $\int_{\gamma_0} \omega = \int_{\gamma_1} \omega$ as desired. □

Problem 2. Consider the holomorphic 1-form

$$\omega = \frac{1}{z^2 - 1} dz.$$

Is this the derivative of a holomorphic function in

- (i) $\mathbb{C} - \{-1, 1\}$?
- (ii) $\mathbb{C} - [-1, 1]$?
- (iii) $\mathbb{C} - (-\infty, 1] - [1, \infty)$?

Explain your answers.

Solution. (i) No. By Cauchy's integral formula applied to the function $f(z) = \frac{1}{z+1}$,

$$\int_{B_{1/2}(1)} \omega = 2\pi i f(1) = \pi i \neq 0,$$

but this integral should be zero if f has an antiderivative.

The answer to both (ii) and (iii) is yes, which are special cases of the following lemma:

Lemma 2.1. Let $\omega = f dz$ be a holomorphic 1-form defined on $U \subseteq \mathbb{C}$ an open subset. Let $z_0 \in U$ be a fixed point. Suppose that for every loop $\gamma: I \rightarrow U$ with $\gamma(0) = \gamma(1) = z_0$,

$$\int_{\gamma} \omega = 0,$$

i.e. line integrals of ω depend only on the endpoints of the path being integrated over. Then f has an antiderivative defined by

$$F(z) = \int_{\gamma_z} \omega$$

where γ_z is an arbitrary path from z_0 to z .

Proof. Let $w_0 \in U$ and pick $\varepsilon > 0$ such that f is given by a convergent power series $f(z) = \sum_{n=0}^{\infty} a_n (z - w_0)^n$ in a neighborhood of $B_{\varepsilon}(w_0) \subseteq U$, and let $\gamma_{w_0, w}: I \rightarrow U$ defined by $\gamma_{w_0, w}(t) = w_0 + (w - w_0)t$ be the line segment from w_0 to w for any $w \in B_{\varepsilon}(w_0)$. Then,

$$\begin{aligned} \frac{F(w) - F(w_0)}{w - w_0} &= \frac{1}{w - w_0} \int_{\gamma_{w_0, w}} \sum_{n=0}^{\infty} a_n (z - w_0)^n dz \\ &= \sum_{n=0}^{\infty} \int_{\gamma_{w_0, w}} a_n (z - w_0)^n \cdot \frac{1}{w - w_0} dz \\ &= \sum_{n=0}^{\infty} \int_0^1 a_n ((w - w_0)t)^n \cdot \frac{1}{w - w_0} \cdot (w - w_0) dt \\ &= \sum_{n=0}^{\infty} \frac{a_n}{n+1} (w - w_0)^{n+1}, \end{aligned}$$

where the sum and integral commute because power series converge locally uniformly. Then, we can take the limit as $w \rightarrow w_0$ to obtain $a_0 = f(w_0)$ (swapping sum and limit by local uniform convergence), so F is indeed holomorphic at every $w_0 \in U$ with derivative f . $\square$

Now we show that the given 1-form ω satisfies the hypotheses of the above lemma in the case of both (ii) and (iii).

(ii) Let γ_1 be the square of side length 1 centered at 1, γ_{-1} the square of side length 1 centered at -1 , and γ the 1×2 rectangle bounding the two squares. By Cauchy's integral formula applied to $\frac{1}{z+1}$ and to $\frac{1}{z-1}$ over γ_1 and γ_{-1} , respectively, we can compute that

$$\int_{\gamma} \omega = \int_{\gamma_1} \omega + \int_{\gamma_{-1}} \omega = 2\pi i \cdot \left(\frac{1}{1+1} + \frac{1}{-1-1} \right) = 0.$$

Let $z_0 = 3/2$ be a point on γ . The fundamental group $\pi_1(\mathbb{C} - [-1, 1], z_0)$ is the infinite cyclic group generated by the homotopy class of γ , so any loop based at z_0 is homotopic to a multiple of γ through a homotopy fixing z_0 ; thus the integral of ω over any loop based at z_0 is zero.

(iii) Any loop based at $z_0 = 0$ is nullhomotopic because $U = \mathbb{C} - (-\infty, -1] - [1, \infty)$ is simply connected; in fact, U is contractible via the homotopy $H: U \times I \rightarrow U$ given by $H(p, t) = tp$. $\square$

Problem 3. Let $\Omega \subseteq \mathbb{C}$ be a good domain and $z_1, \dots, z_k \in \text{Int } \Omega$. Suppose that $f: \Omega - \{z_1, \dots, z_k\} \rightarrow \mathbb{C}$ is holomorphic and satisfies

$$\lim_{z \rightarrow z_j} (z - z_j) f(z) = 0$$

for $j = 1, \dots, k$. Show that

$$f(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} d\zeta$$

for $z \in \Omega - \{z_1, \dots, z_k\}$.

Solution. Given $j \in \{1, \dots, k\}$, let $\varepsilon > 0$ be such that $B_\varepsilon(z_j)$ is contained in the interior of Ω and does not contain any other z_i . On this open ball, define $g_j(z) = (z - z_j)^2 f(z)$ for $z \neq z_j$ and $g_j(z_j) = 0$. This is of course holomorphic on $B_\varepsilon(z_j) - \{z_j\}$ and

$$\lim_{z \rightarrow z_j} \frac{g_j(z) - g_j(z_j)}{z - z_j} = \lim_{z \rightarrow z_j} (z - z_j) f(z) = 0,$$

so g_j is holomorphic on $B_\varepsilon(z_j)$. Thus g_j has a power series representation $g_j(z) = \sum_{n=0}^{\infty} a_n (z - z_j)^n$. Since $g_j(z_j) = 0$, $a_0 = 0$, and $g'_j(z_j) = 0$, so $a_1 = 0$. Thus

$$\frac{g_j(z)}{(z - z_j)^2} = \sum_{n=0}^{\infty} a_{n+2} (z - z_j)^n$$

equals $f(z)$ on $B_\varepsilon(z_j) - \{z_j\}$, meaning f extends to a holomorphic function $g_j(z)$ on the entirety of $B_\varepsilon(z_j)$. We have therefore shown that there exists a holomorphic function $\tilde{f}: \Omega \rightarrow \mathbb{C}$ such that $\tilde{f}(z) = f(z)$ for $z \in \Omega - \{z_1, \dots, z_k\}$, so, by Cauchy's integral formula,

$$\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(z)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\tilde{f}(z)}{\zeta - z} d\zeta = \tilde{f}(z) = f(z)$$

for all $z \in \Omega - \{z_1, \dots, z_k\}$. $\square$